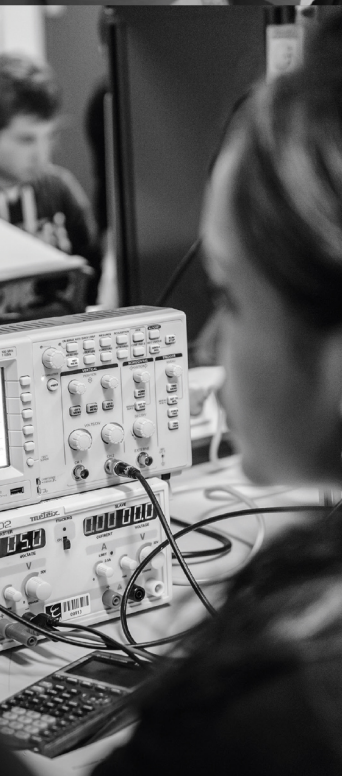




ENSEA

Beyond Engineering

ELECTRONIQUE DE PUISSANCE

DEE_2401

TD

Alexis MARTIN
2026

1 Alimentation d'un microprocesseur

On souhaite alimenter un microprocesseur STM32H4 à l'aide d'une batterie. Le cahier des charges est le suivant:

- Batterie LiPo 2S $V_{in} = 7.4\text{ V}$
- Tension d'alimentation du microprocesseur: $V_{out} = 3.3\text{ V}$
- Ondulation de tension max en entrée du microprocesseur: $\Delta V_S = 50\text{ mV}$
- Courant nominal de sortie: $I_S = 200\text{ mA}$
- Ondulation de courant: 30%

Pour cela, on souhaite utiliser le composant suivant: **LT1959** dont un extrait de la documentation est en annexe. Afin de dimensionner tous les composants, on décide d'utiliser des IAG: Claude (Sonnet 4.6), Copilot (M365 Copilot basé sur GPT-5.5 Automatique) et ChatGPT (GPT 5.5 Instant). Les réponses des IAs ont été générées en juin 2026. Voici le Prompt utilisé:

"Je souhaite utiliser le composant LT1959, pour alimenter un STM32H4 à l'aide d'une batterie. Voici le cahier des charges:

- Batterie LiPo 2S $V_{in} = 7.4\text{ V}$
- Tension d'alimentation du microprocesseur: $V_{out} = 3.3\text{ V}$
- Ondulation de tension max en entrée du microprocesseur: $\Delta V_S = 50\text{ mV}$
- Courant nominal de sortie: $I_S = 200\text{ mA}$
- Ondulation de courant: 30%

Dans un fichier word, détaille moi les calculs pour le dimensionnement des composants passifs. Ajoute une image du schéma électrique. Ajoute une autre image avec la forme du courant dans l'inductance, de la tension sur la broche SW et de la tension de sortie".

A partir des réponses des IAs et de vos connaissances:

1. Représenter le schéma électrique du convertisseur Buck
2. Donner les formes d'ondes de la tension de sortie, de la tension sur la broche SW et du courant dans l'inductance
3. Dimensionner les composants passifs du convertisseur Buck

Copilot

1. Cahier des charges

$V_{in} = 7.4 \text{ V}$ (LiPo 2S)

$V_{out} = 3.3 \text{ V}$

$I_{out} = 200 \text{ mA}$

Ondulation courant $\Delta I_L = 30\%$

Ondulation tension $\Delta V_{out} \text{ max} = 50 \text{ mV}$

2. Rapport cyclique

$D = V_{out} / V_{in} = 3.3 / 7.4 = 0.446$

3. Dimensionnement de l inductance

Formule : $L = V_{out} * (V_{in} - V_{out}) / (V_{in} * \Delta I_L * f_s)$

$\Delta I_L = 0.3 \times 0.2 = 0.060 \text{ A}$

$L = 60.95 \text{ } \mu\text{H} \rightarrow \text{valeur normalisée : } 68 \text{ } \mu\text{H}$

Courant maximal = $0.230 \text{ A} \rightarrow \text{choisir } > 0.5 \text{ A}$

4. Condensateur de sortie

Formule : $C_{out} = \Delta I_L / (8 \times f_s \times \Delta V_{out})$

$C_{out} = 0.30 \text{ } \mu\text{F} \rightarrow \text{choix : } 47 \text{ } \mu\text{F} \text{ céramique faible ESR}$

5. Condensateur d entrée

Choix recommandé : $10 \text{ } \mu\text{F}$ à $22 \text{ } \mu\text{F}$ céramique

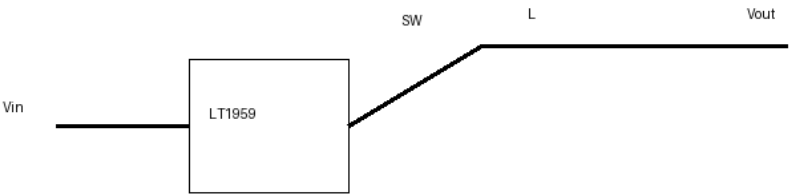
6. Pont diviseur

$V_{out} = V_{ref} (1 + R_1/R_2)$

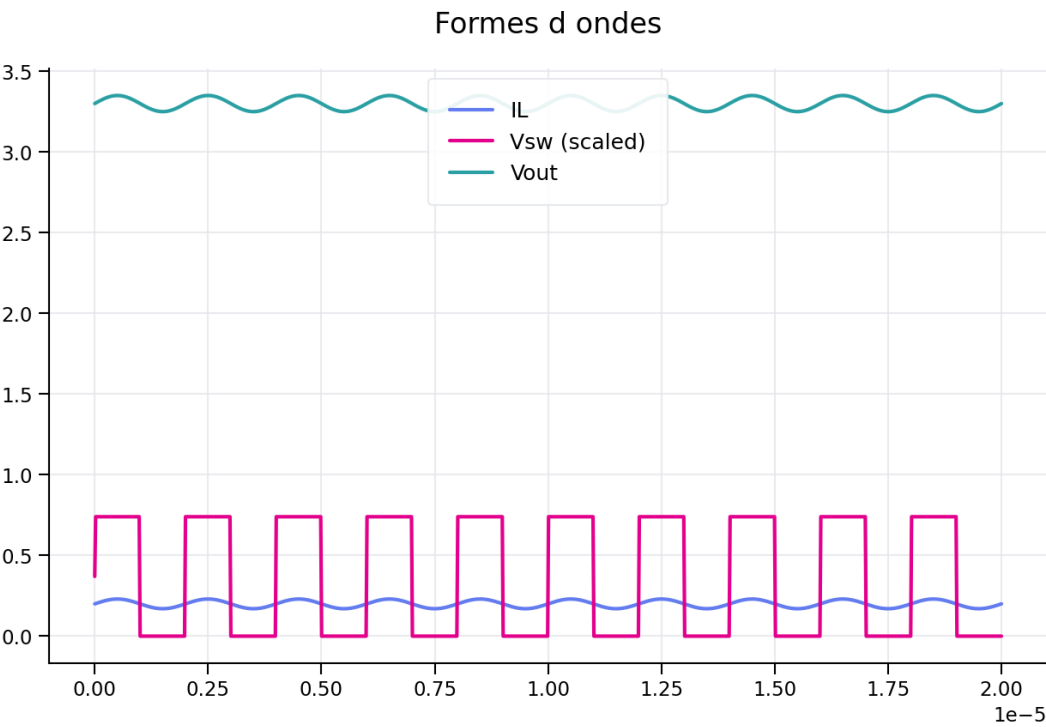
$R_2 = 10 \text{ k}\Omega \rightarrow R_1 = 16.83 \text{ k}\Omega \rightarrow \text{valeur normalisée : } 16.9 \text{ k}\Omega$

7. Schéma électrique

Schéma Buck LT1959



8. Formes d ondes



Claude

Paramètre	Valeur
Tension d'entrée (batterie LiPo 2S)	VIN = 7,4 V
Tension de sortie	VOUT = 3,3 V
Courant nominal de sortie	IS = 200 mA
Ondulation courant inductance	$\Delta IL = 30 \% \text{ de IS}$
Ondulation tension sortie max	$\Delta VS = 50 \text{ mV}$
Fréquence de découpage (LT1959)	f = 500 kHz

1. Rapport cyclique (Duty Cycle)

Pour un convertisseur buck idéal en mode continu, le rapport cyclique est :

$$D = V_{OUT} / V_{IN}$$

$$D = 3,3 / 7,4$$

$$D = 0,446 \rightarrow \text{ton} = 0,892 \mu\text{s} / \text{toff} = 1,108 \mu\text{s}$$

Vérification : $D = 44,6 \% < 50 \% \rightarrow$ courant limite IP = 4,5 A garanti par le LT1959.

2. Calcul de l'inductance L1

2.1 Ondulation de courant cible

L'ondulation crête-à-crête de courant dans l'inductance est fixée à 30 % du courant de sortie :

$$\Delta IL = 0,30 \times IS = 0,30 \times 200 \text{ mA}$$

$$\Delta IL = 60 \text{ mA}$$

2.2 Valeur minimale de L

La formule issue de la datasheet LT1959 pour le mode continu est :

$$L = (V_{OUT} \times (V_{IN} - V_{OUT})) / (V_{IN} \times \Delta IL \times f)$$

$$L = (3,3 \times (7,4 - 3,3)) / (7,4 \times 0,060 \times 500 \times 10^3)$$

$$L = (3,3 \times 4,1) / (7,4 \times 30\,000)$$

$$L = 13,53 / 222\,000$$

$$L_{\min} = 60,9 \mu\text{H}$$

Valeur normalisée choisie : L1 = 10 μH (série standard, disponible avec Isat > 500 mA).

Remarque : avec L = 10 μH , l'ondulation réelle sera plus faible que la cible de 30 % :

$$\Delta IL_{\text{réel}} = (V_{OUT} \times (V_{IN} - V_{OUT})) / (V_{IN} \times L \times f)$$

$$\Delta IL_{\text{réel}} = (3,3 \times 4,1) / (7,4 \times 10 \times 10^{-6} \times 500 \times 10^3)$$

$$\Delta IL_{\text{réel}} \approx 365 \text{ mA} \rightarrow 18 \% \text{ de IS } \checkmark (< 30 \%)$$

3. Courants crête et vallée dans l'inductance

$$IL_{\text{peak}} = IS + \Delta IL_{\text{réel}}/2 = 200 + 183 = 383 \text{ mA}$$

$$IL_{\text{valley}} = IS - \Delta IL_{\text{réel}}/2 = 200 - 183 = 17 \text{ mA}$$

Vérification mode continu : $IL_{\text{valley}} > 0 \checkmark$

Courant de saturation requis : Isat > $IL_{\text{peak}} = 383 \text{ mA} \rightarrow$ inductor standard 10 μH suffisant.

4. Condensateur de sortie C3

4.1 Contrainte ESR (ondulation de tension)

L'ondulation de tension en sortie est principalement déterminée par l'ESR du condensateur :

$$\begin{aligned}\Delta V_S &\approx \Delta I_{L_réel} \times ESR \\ ESR_{max} &= \Delta V_S / \Delta I_{L_réel} = 50 \times 10^{-3} / 0,365 \\ ESR_{max} &= 136 \text{ m}\Omega\end{aligned}$$

Condensateur choisi : AVX TPS 100 μ F / 10 V (tantalum solide, ESR typ. 100 m Ω < 136 m Ω) ✓

4.2 Valeur en capacité

La valeur en μ F n'est pas critique. Avec C3 = 100 μ F :

$$\begin{aligned}\text{Ondulation capacitive} &= \Delta I_L / (8 \times f \times C3) \\ &= 0,365 / (8 \times 500 \times 10^3 \times 100 \times 10^{-6}) \\ &= 0,91 \text{ mV (négligeable devant la contribution ESR)}\end{aligned}$$

5. Condensateur d'entrée C1

Le courant RMS dans le condensateur d'entrée est :

$$\begin{aligned}I_{C1_rms} &= I_S \times \sqrt{(V_{OUT} \times (V_{IN} - V_{OUT})) / V_{IN}} \\ I_{C1_rms} &= 0,200 \times \sqrt{(3,3 \times 4,1 / 7,4)} \\ I_{C1_rms} &= 0,200 \times \sqrt{0,183} \\ I_{C1_rms} &\approx 85,5 \text{ mA}\end{aligned}$$

Condensateur choisi : C1 = 100 μ F / 10 V, tantalum AVX TPS (Iripple $\geq$ 0,7 A) ✓

6. Diode de roue libre D1

Courant moyen dans D1 :

$$\begin{aligned}I_{D1_avg} &= I_S \times (V_{IN} - V_{OUT}) / V_{IN} \\ I_{D1_avg} &= 0,200 \times (7,4 - 3,3) / 7,4 \\ I_{D1_avg} &= 110 \text{ mA}\end{aligned}$$

Tension inverse maximale = $V_{IN} = 7,4$ V

Diode choisie : MBRS330T3 (Schottky, 3 A / 30 V, V_F typ. 0,5 V @ 3 A) ✓

7. Condensateur BOOST C2

Le condensateur BOOST alimente le driver de l'interrupteur NPN pour le saturer. La valeur minimale est :

$$\begin{aligned}C2_{min} &= (I_S / 50) / (f \times 0,6) \\ C2_{min} &= (0,200 / 50) / (500 \times 10^3 \times 0,6) \\ C2_{min} &\approx 13 \text{ nF} \rightarrow C2 \text{ choisi} = 0,68 \text{ }\mu\text{F (valeur datasheet standard)} \checkmark\end{aligned}$$

Diode D2 : 1N914 ou 1N4148, anode connectée à VOUT.

8. Réseau de contre-réaction (R1, R2)

La tension de référence interne est $V_{REF} = 1,21$ V. Le diviseur résistif R1/R2 fixe VOUT :

$$\begin{aligned}V_{OUT} &= V_{REF} \times (1 + R1/R2) \\ \rightarrow R1/R2 &= (V_{OUT}/V_{REF}) - 1 = (3,3/1,21) - 1 = 1,727\end{aligned}$$

Choix de $R2 \leq 2,5$ k Ω (recommandation datasheet) : $R2 = 2,49$ k Ω

$$R1 = R2 \times (V_{OUT}/V_{REF} - 1) = 2490 \times 1,727$$

$R1 = 4\,298\,\Omega \rightarrow$ valeur normalisée : $R1 = 4,32\,\text{k}\Omega$ (E96, 1 %)

Vérification :

$$V_{OUT} = 1,21 \times (1 + 4320/2490) = 1,21 \times 2,735$$

$$V_{OUT} = 3,309\,\text{V} \checkmark \text{ (erreur } < 0,3\,\% \text{)}$$

9. Condensateur de compensation VC (CC)

La compensation par défaut recommandée par la datasheet pour la majorité des applications est :

$$CC = 1,5\,\text{nF} \text{ (condensateur seul, } RC = 0)$$

Ce réseau crée un pôle à :

$$f_{\text{pole}} = 1 / (2\pi \times R_{OUT_EA} \times CC) \approx 530\,\text{Hz}$$

Le déphasage de boucle reste $> 60^\circ$ à la fréquence de coupure ($\approx 100\,\text{kHz}$), garantissant la stabilité.

10. Bilan des composants

Référence	Valeur / Type	Rôle	Critère de choix
L1	10 μH , Isat $> 400\,\text{mA}$	Inductance de filtrage	$\Delta IL \approx 18\,\%$
C1	100 μF / 10 V Tant.	Condensateur entrée	$I_{\text{ripple}} \geq 86\,\text{mA}$
C2	0,68 μF céramique	Condensateur BOOST	Charge driver NPN
C3	100 μF / 10 V Tant.	Condensateur sortie	$ESR \leq 136\,\text{m}\Omega$
CC	1,5 nF céramique	Compensation VC	Stabilité boucle
D1	MBRS330T3 (3A/30V)	Diode roue libre	VF faible, Schottky
D2	1N914 / 1N4148	Diode BOOST	Rapide, faible VF
R1	4,32 $\text{k}\Omega$ 1 %	Diviseur FB (haut)	Réglage V_{OUT}
R2	2,49 $\text{k}\Omega$ 1 %	Diviseur FB (bas)	Réglage V_{OUT}

11. Schéma électrique

Schéma électrique — Convertisseur Buck LT1959 (7.4V→3.3V / 200mA)

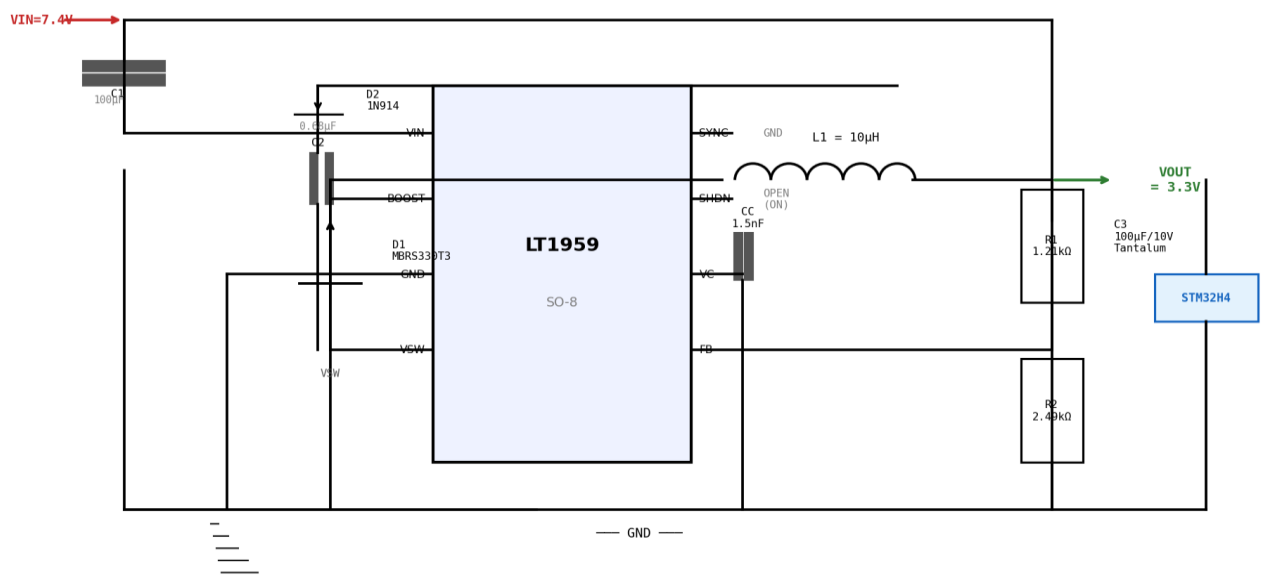


Figure 1 — Schéma de principe du convertisseur Buck LT1959 (7,4 V → 3,3 V / 200 mA)

12. Formes d'ondes simulées

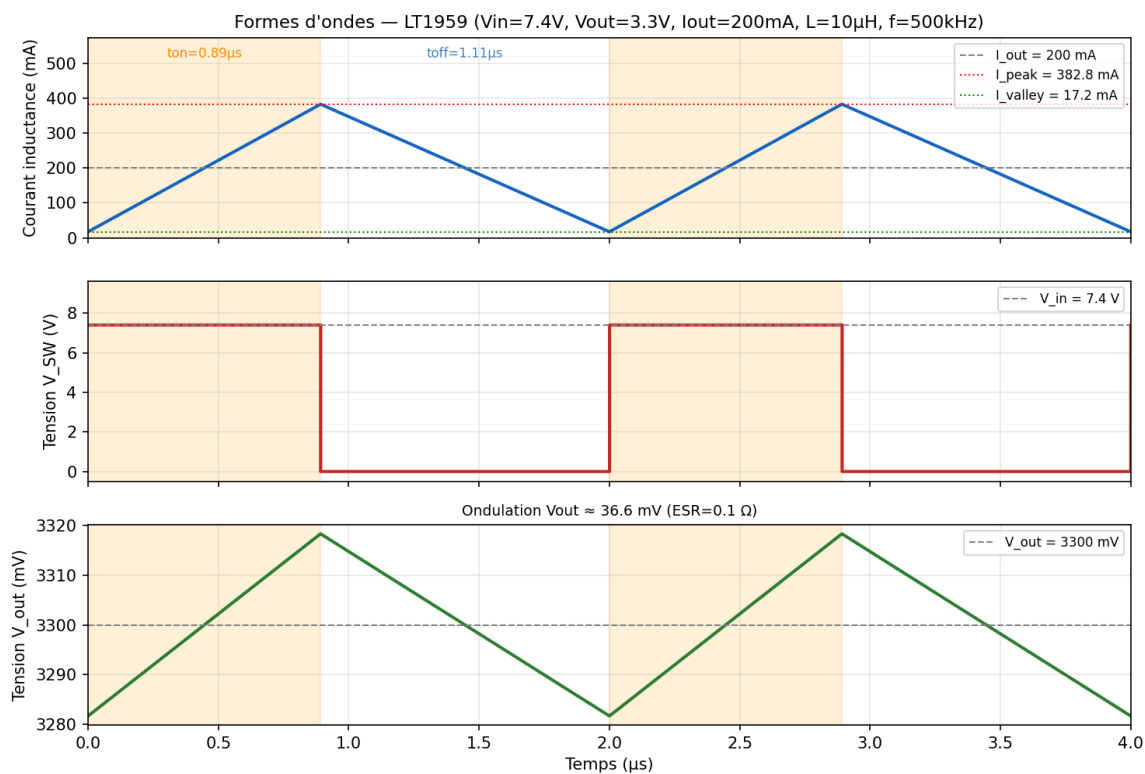


Figure 2 — Courant inductance I_L , tension V_{SW} et tension de sortie V_{OUT} sur 2 périodes ($f = 500 \text{ kHz}$)

La zone orangée représente le temps de conduction de l'interrupteur ($t_{on} \approx 0,892 \mu s$).

L'ondulation de V_{OUT} est principalement due à l'ESR du condensateur C3 ($\approx 36 \text{ mV} < 50 \text{ mV spec}$) ✓

Référence : Linear Technology — LT1959 Datasheet, Rev. A.

Calculs réalisés pour $I_S = 200 \text{ mA}$ nominal. Vérifier les paramètres thermiques pour des courants supérieurs.

ChatGPT

Objet : dimensionner les composants passifs principaux d'un convertisseur abaisseur LT1959 pour générer 3,3 V / 200 mA à partir d'une batterie LiPo 2S nominale de 7,4 V, avec une ondulation de sortie maximale demandée de 50 mV et une ondulation de courant inductance de 30 %.

1. Données et hypothèses

Grandeur	Valeur	Commentaire
Vin	7,4 V	Tension nominale batterie LiPo 2S
Vout	3,3 V	Alimentation microprocesseur
Is	200 mA	Courant nominal
ΔVS max	50 mV crête-à-crête	Ondulation maximale admise en entrée du STM32
ΔIL	30 % de Is = 60 mA c-à-c	Ondulation de courant inductance
fs	500 kHz	Fréquence fixe du LT1959
VFB	1,21 V	Référence de la broche FB

Remarque importante : les calculs ci-dessous sont réalisés au point nominal $V_{in} = 7,4\text{ V}$. Pour une conception de série, il faut également vérifier la batterie pleine charge, typiquement 8,4 V pour une LiPo 2S, et la batterie déchargée. Le LT1959 étant largement surdimensionné en courant pour 200 mA, les contraintes dominantes seront plutôt l'ondulation, la stabilité, le choix du condensateur de sortie et le routage.

2. Schéma électrique de principe

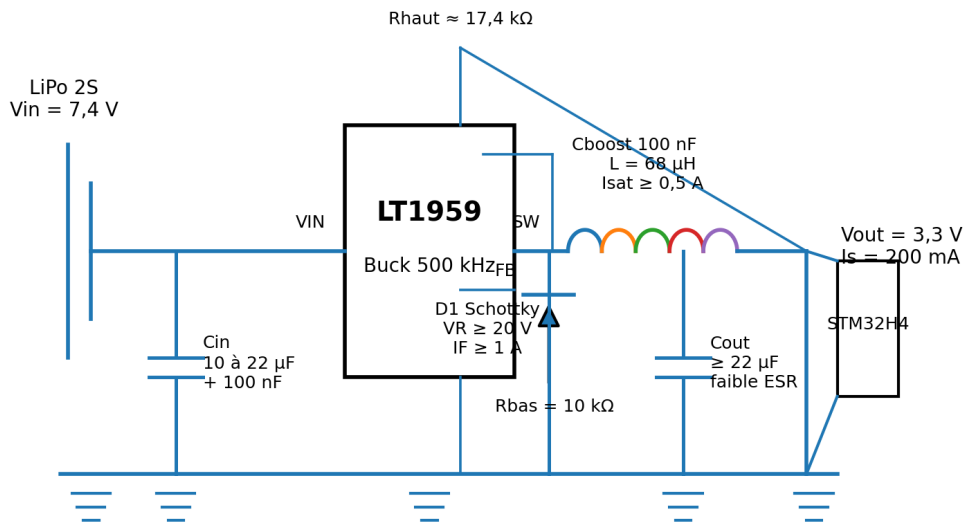


Schéma de principe pour le dimensionnement. Les valeurs finales doivent être validées avec la compensation, le routage et la fiche technique.

Figure 1 - Schéma de principe du convertisseur buck LT1959 vers 3,3 V.

3. Rapport cyclique

En première approximation, en négligeant les chutes dans l'interrupteur interne et dans la diode de roue libre :

$D = V_{out} / V_{in} = 3,3 / 7,4 = 0.446$, soit 44.6 %.

À 500 kHz, la période de découpage vaut $T = 1 / f_s = 2.00 \mu s$. On obtient $T_{on} = D.T = 0.89 \mu s$ et $T_{off} = (1-D).T = 1.11 \mu s$.

4. Dimensionnement de l'inductance

Pour un buck en conduction continue, l'ondulation crête-à-crête du courant d'inductance est :

$$\Delta I_L = (V_{in} - V_{out}).D / (L.f_s)$$

$$\text{Donc } L = (V_{in} - V_{out}).D / (f_s.\Delta I_L) = (7,4 - 3,3) \times 0.446 / (500 \text{ kHz} \times 60 \text{ mA}) = 60.9 \mu H.$$

Valeur normalisée conseillée : 68 μH . Avec 68 μH , l'ondulation vaut environ 53.8 mA c-à-c, soit environ 26.9 % du courant nominal.

Courant crête dans l'inductance : $I_{pk} = I_s + \Delta I_L/2 = 200 \text{ mA} + 30 \text{ mA} = 230 \text{ mA}$. Courant minimal : $I_{min} = 170 \text{ mA}$. Choisir une inductance avec courant de saturation nettement supérieur, par exemple $I_{sat} \geq 0,5 \text{ A}$, et une résistance série faible pour limiter les pertes à faible tension.

5. Condensateur de sortie

L'ondulation de sortie possède deux contributions principales : la charge/décharge capacitive et la chute ESR du condensateur. La composante capacitive idéale est :

$$\Delta V_{cap} = \Delta I_L / (8.f_s.C_{out})$$

$$\text{En imposant } \Delta V_{cap} \leq 50 \text{ mV} : C_{out} \geq \Delta I_L / (8.f_s.\Delta V) = 60 \text{ mA} / (8 \times 500 \text{ kHz} \times 50 \text{ mV}) = 0.30 \mu F.$$

La contrainte purement capacitive est donc très faible. La contrainte ESR donne : $ESR \leq \Delta V / \Delta I_L = 50 \text{ mV} / 60 \text{ mA} = 0.83 \Omega$.

Choix pratique recommandé : au minimum 22 μF céramique X7R en sortie, complété si nécessaire par 100 nF proche de la charge. La valeur réelle d'un condensateur céramique baisse avec la polarisation DC : il faut donc vérifier la capacité effective à 3,3 V. Pour un microprocesseur, l'objectif n'est pas seulement l'ondulation à 500 kHz, mais aussi la tenue aux appels de courant rapides.

6. Condensateur d'entrée

Le condensateur d'entrée doit fournir le courant pulsé du convertisseur et rester proche des broches VIN/GND du LT1959. Son courant RMS approximatif vaut :

$$I_{cin,rms} \approx I_s.\sqrt{D.(1-D)} = 0,2 \times \sqrt{0.446 \times 0.554} = 99 \text{ mA RMS}.$$

Pour viser par exemple 100 mV d'ondulation d'entrée basse fréquence de découpage, une estimation donne $C_{in} \geq I_s.D.(1-D)/(f_s.\Delta V_{in}) = 0.99 \mu F$. En pratique, retenir 10 à 22 μF faible ESR, avec un 100 nF céramique directement au plus près de VIN et GND.

7. Pont de retour FB

La tension de sortie est fixée par : $V_{out} = V_{FB} \times (1 + R_{haut}/R_{bas})$.

$$\text{En choisissant } R_{bas} = 10 \text{ k}\Omega : R_{haut} = 10 \text{ k}\Omega \times (3,3/1,21 - 1) = 17.27 \text{ k}\Omega.$$

Valeur normalisée : $R_{haut} = 17,4 \text{ k}\Omega$ et $R_{bas} = 10 \text{ k}\Omega$, ce qui donne $V_{out} \approx 3,32 \text{ V}$. Une alternative est $R_{haut} = 17,2 \text{ k}\Omega$ pour obtenir environ 3,29 V.

8. Diode de roue libre et condensateur boost

La diode de roue libre conduit pendant T_{off} . Son courant moyen vaut $I_{d,moy} \approx I_s \cdot (1-D) = 0,2 \times 0,554 = 111 \text{ mA}$. Son courant crête est proche du courant crête inductance, soit environ 230 mA au point nominal.

Choisir une diode Schottky rapide avec $V_{RRM} \geq 20 \text{ V}$ et $I_F \geq 1 \text{ A}$ pour disposer d'une marge confortable. La tension inverse doit au minimum supporter V_{in} , et le routage doit minimiser la boucle LT1959 - diode - condensateur d'entrée.

Le condensateur boost utilisé par la commande interne est typiquement de 100 nF, placé au plus près des broches BOOST et SW, selon le schéma d'application de la fiche technique.

9. Formes d'ondes attendues

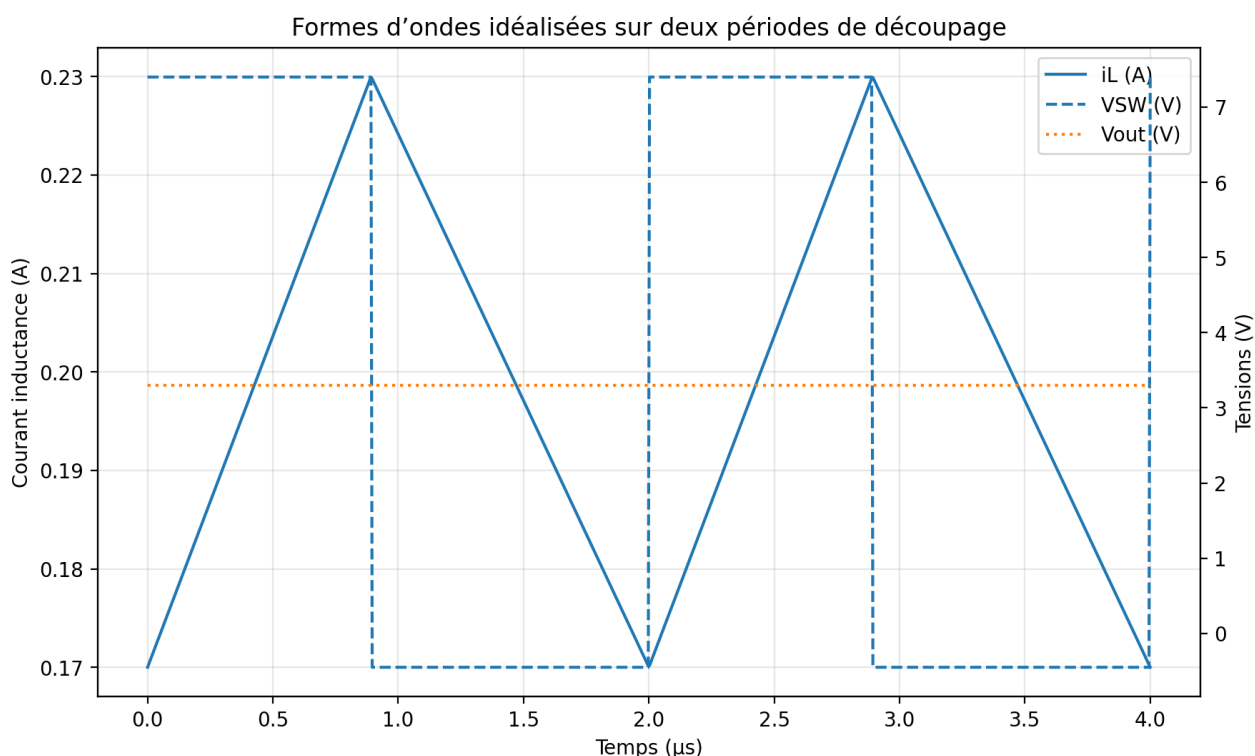


Figure 2 - Courant dans l'inductance, tension SW et ondulation de sortie idéalisés.

Le courant inductance est triangulaire en conduction continue. La broche SW commute entre une valeur proche de V_{in} pendant T_{on} et une tension légèrement négative pendant T_{off} , limitée par la diode de roue libre. La tension de sortie est quasi continue, avec une ondulation résiduelle dépendant de ΔI_L , C_{out} , de l'ESR et du routage.

10. Synthèse des valeurs proposées

Composant	Valeur calculée / contrainte	Choix proposé
L	60,9 μH pour $\Delta\text{IL} = 60 \text{ mA}$	68 μH , $\text{Isat} \geq 0,5 \text{ A}$
Cout	$\geq 0,30 \mu\text{F}$ et $\text{ESR} \leq 0,83 \Omega$	22 μF X7R + 100 nF, faible ESR
Cin	$\text{Irms} \approx 99 \text{ mA}$; capacité minimale faible	10 à 22 μF faible ESR + 100 nF proche LT1959
Rhaut/Rbas	$\text{Rhaut} = 17,27 \text{ k}\Omega$ si $\text{Rbas} = 10 \text{ k}\Omega$	17,4 $\text{k}\Omega$ / 10 $\text{k}\Omega$
D1	$\text{Id,moy} \approx 111 \text{ mA}$; $\text{Icrête} \approx 230 \text{ mA}$	Schottky $\geq 20 \text{ V}$, $\geq 1 \text{ A}$
Cboost	Selon application LT1959	100 nF proche BOOST-SW

Conclusion : au point nominal 7,4 V vers 3,3 V / 200 mA, une inductance de 68 μH et un condensateur de sortie céramique de 22 μF donnent une marge confortable vis-à-vis de l'ondulation de 50 mV. Il faut toutefois valider la stabilité de boucle avec la compensation de la broche VC, les caractéristiques réelles des condensateurs et le routage du circuit imprimé.

Références

Analog Devices, LT1959 product page et fiche technique : régulateur abaisseur 500 kHz, sortie ajustable par FB 1,21 V, interrupteur 4,5 A, entrée minimale 4 V, recommandations de routage VIN/GND/SW/diode.

4.5A, 500kHz Step-Down Switching Regulator

FEATURES

- Operates with Input as Low as 4V
- Output Range Down to 1.21V
- Constant 500kHz Switching Frequency
- Uses All Surface Mount Components
- Inductor Size Reduced to 1.8μH
- Saturating Switch Design: 0.07Ω
- Shutdown Current: 15μA
- Easily Synchronizable
- Cycle-by-Cycle Current Limiting
- 4.5A Switch
- Current Mode Control

APPLICATIONS

- Portable Computers
- Battery-Powered Systems
- Battery Chargers
- Distributed Power
- 5V to 3.3V Conversion
- 5V to 2.5V Conversion
- 5V to 1.8V Conversion

DESCRIPTION

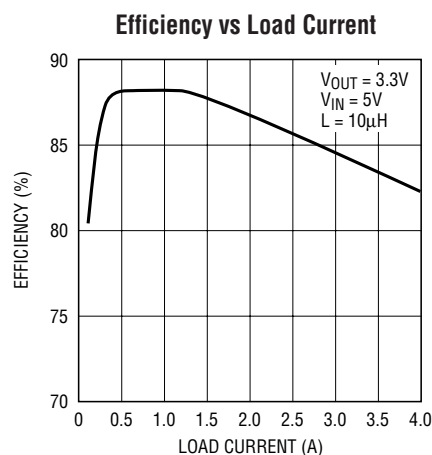
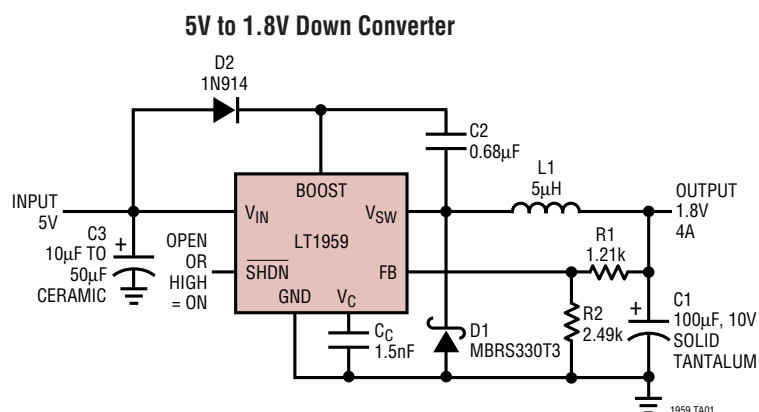
The LT[®]1959 is a 500kHz monolithic buck mode switching regulator functionally identical to the LT1506 but optimized for lower output voltage applications. It will operate down to 1.21V output compared to 2.42V for the LT1506. A 4.5A switch is included on the die along with all the necessary oscillator, control and logic circuitry. High switching frequency allows a considerable reduction in the size of external components. The topology is current mode for fast transient response and good loop stability.

A special high speed bipolar process and new design techniques achieve high efficiency at high switching frequency. Efficiency is maintained over a wide output current range by keeping quiescent supply current to 3.8mA and by utilizing a supply boost capacitor to saturate the power switch.

The LT1959 fits into standard 7-pin DD and fused lead SO-8 packages. Full cycle-by-cycle short-circuit protection and thermal shutdown are provided. Standard surface mount external parts are used, including the inductor and capacitors. There is the optional function of shutdown or synchronization. A shutdown signal reduces supply current to 15μA. Synchronization allows an external logic level signal to increase the internal oscillator from 580kHz to 1MHz.

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TYPICAL APPLICATION

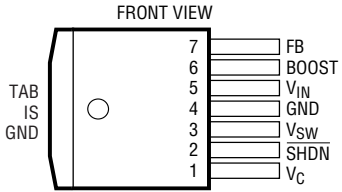
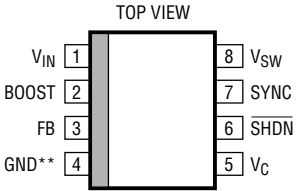


1959 TA02

ABSOLUTE MAXIMUM RATINGS (Note 1)

Input Voltage	16V	SYNC Pin Voltage	7V
BOOST Voltage	30V	Operating Junction Temperature Range	
BOOST Pin Above Input Voltage	15V	LT1959C	0°C to 125°C
SHDN Pin Voltage	7V	LT1959I	–40°C to 125°C
FB Pin Voltage	3.5V	Storage Temperature Range	–65°C to 150°C
FB Pin Current	1mA	Lead Temperature (Soldering, 10 sec)	300°C

PACKAGE/ORDER INFORMATION

 FRONT VIEW R PACKAGE 7-LEAD PLASTIC DD $T_{JMAX} = 150^{\circ}\text{C}$, $\theta_{JA} = 30^{\circ}\text{C/W}$ WITH PACKAGE SOLDERED TO 0.5 SQUARE INCH COPPER AREA OVER BACKSIDE GROUND PLANE OR INTERNAL POWER PLANE. θ_{JA} CAN VARY FROM 20°C/W TO >40°C/W DEPENDING ON MOUNTING TECHNIQUES	ORDER PART NUMBER	 TOP VIEW S8 PACKAGE 8-LEAD PLASTIC SO $T_{JMAX} = 150^{\circ}\text{C}$, $\theta_{JA} = 80^{\circ}\text{C/W}$ ** WITH (FUSED GND) GROUND PIN CONNECTED TO GROUND PLANE OR LARGE LANDS	ORDER PART NUMBER
	LT1959CR LT1959IR		LT1959CS8 LT1959IS8
			S8 PART MARKING
			1959 1959I

ELECTRICAL CHARACTERISTICS The ● denotes the specifications which apply over the full operating temperature range, otherwise specifications are at $T_A = 25^{\circ}\text{C}$, $T_J = 25^{\circ}\text{C}$, $V_{IN} = 5\text{V}$, $V_C = 1.5\text{V}$, $\text{Boost} = V_{IN} + 5\text{V}$, switch open, unless otherwise noted.

PARAMETER	CONDITIONS		MIN	TYP	MAX	UNITS
Feedback Voltage (Adjustable)	All Conditions	●	1.19	1.21	1.23	V
Reference Voltage Line Regulation	$4.3\text{V} \leq V_{IN} \leq 15\text{V}$			0.01	0.03	%/V
Feedback Input Bias Current		●	–0.5	0	0.5	μA
Error Amplifier Voltage Gain	(Note 2)		200	400		
Error Amplifier Transconductance	$\Delta I(V_C) = \pm 10\mu\text{A}$	●	1500 1000	2000	2700 3100	μMho μMho
V_C Pin to Switch Current Transconductance				5.3		A/V
Error Amplifier Source Current	$V_{FB} = 1.05\text{V}$	●	140	225	320	μA
Error Amplifier Sink Current	$V_{FB} = 1.35\text{V}$	●	140	225	320	μA
V_C Pin Switching Threshold	Duty Cycle = 0			0.9		V
V_C Pin High Clamp				2.1		V
Switch Current Limit	V_C Open, $V_{FB} = 1.05\text{V}$, DC $\leq 50\%$	●	4.5	6	8.5	A
Slope Compensation	DC = 80%			0.8		A

ELECTRICAL CHARACTERISTICS

The ● denotes the specifications which apply over the full operating temperature range, otherwise specifications are at $T_A = 25^\circ\text{C}$, $T_J = 25^\circ\text{C}$, $V_{IN} = 5\text{V}$, $V_C = 1.5\text{V}$, $\text{Boost} = V_{IN} + 5\text{V}$, switch open, unless otherwise noted.

PARAMETER	CONDITIONS		MIN	TYP	MAX	UNITS
Switch On Resistance (Note 7)	$I_{SW} = 4.5\text{A}$	●		0.07	0.1 0.13	Ω Ω
Maximum Switch Duty Cycle	$V_{FB} = 1.05\text{V}$	●	90 86	93 93		% %
Switch Frequency	V_C Set to Give 50% Duty Cycle	●	460 440	500	540 560	kHz kHz
Switch Frequency Line Regulation	$4.3\text{V} \leq V_{IN} \leq 15\text{V}$	●		0	0.15	%/V
Frequency Shifting Threshold on FB Pin	$\Delta f = 10\text{kHz}$	●	0.5	0.7	1.0	V
Minimum Input Voltage (Note 3)		●		4.0	4.3	V
Minimum Boost Voltage (Note 4)	$I_{SW} \leq 4.5\text{A}$	●		2.3	3.0	V
Boost Current (Note 5)	$I_{SW} = 1\text{A}$	●		20	35	mA
	$I_{SW} = 4.5\text{A}$	●		90	140	mA
Input Supply Current (Note 6)		●		3.8	5.4	mA
Shutdown Supply Current	$V_{SHDN} = 0\text{V}$, $V_{SW} = 0\text{V}$, V_C Open	●		15	50 75	μA μA
Lockout Threshold	V_C Open	●	2.3	2.38	2.46	V
Shutdown Thresholds	V_C Open Device Shutting Down	●	0.13	0.37	0.60	V
	Device Starting Up	●	0.25	0.45	0.7	V
Synchronization Threshold		●		1.5	2.2	V
Synchronizing Range			580		1000	kHz
SYNC Pin Input Resistance				40		$\text{k}\Omega$

Note 1: Absolute Maximum Ratings are those values beyond which the life of a device may be impaired.

Note 2: Gain is measured with a V_C swing equal to 200mV above the switching threshold level to 200mV below the upper clamp level.

Note 3: Minimum input voltage is not measured directly, but is guaranteed by other tests. It is defined as the voltage where internal bias lines are still regulated so that the reference voltage and oscillator frequency remain constant. Actual minimum input voltage to maintain a regulated output will depend on output voltage and load current. See Applications Information.

Note 4: This is the minimum voltage across the boost capacitor needed to guarantee full saturation of the internal power switch.

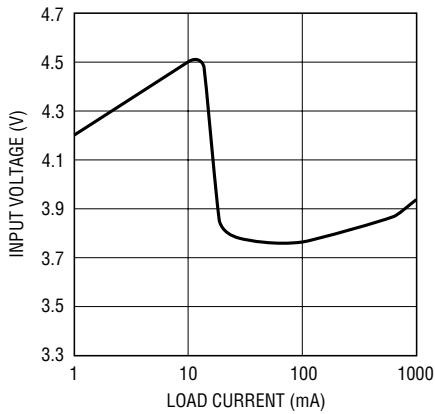
Note 5: Boost current is the current flowing into the boost pin with the pin held 5V above input voltage. It flows only during switch on time.

Note 6: Input supply current is the bias current drawn by the input pin with switching disabled.

Note 7: Switch on resistance is calculated by dividing V_{IN} to V_{SW} voltage by the forced current (4.5A). See Typical Performance Characteristics for the graph of switch voltage at other currents.

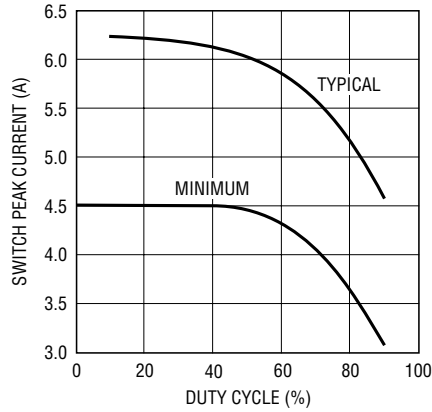
TYPICAL PERFORMANCE CHARACTERISTICS

**Minimum Input Voltage
with 3.3V Output**



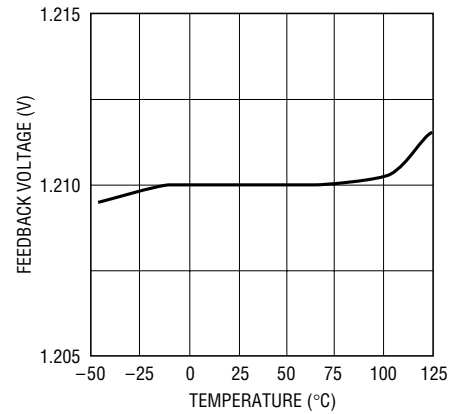
1959 G01

Switch Peak Current Limit



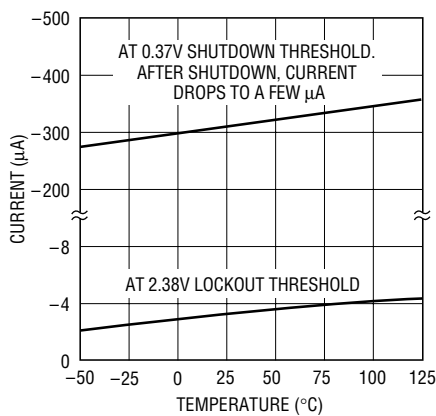
1959 G02

Feedback Pin Voltage



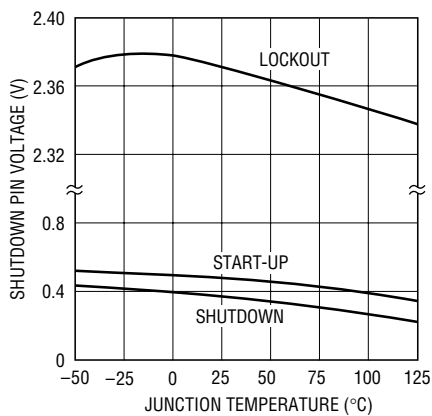
1959 G03

Shutdown Pin Bias Current



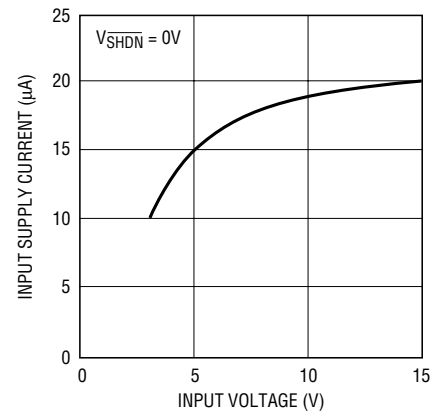
1959 G04

**Lockout and Shutdown
Thresholds**



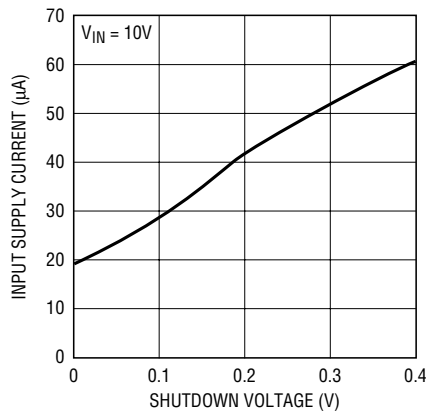
1959 G05

Shutdown Supply Current



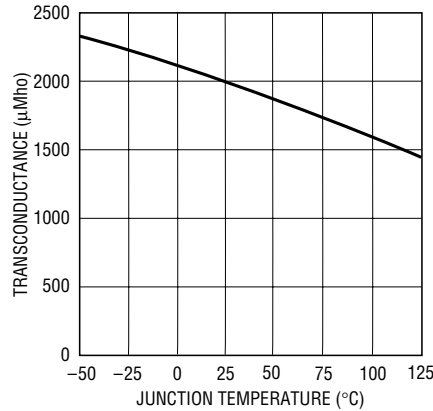
1959 G06

Shutdown Supply Current



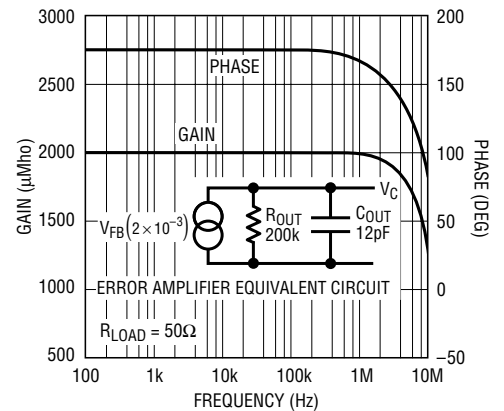
1959 G07

Error Amplifier Transconductance



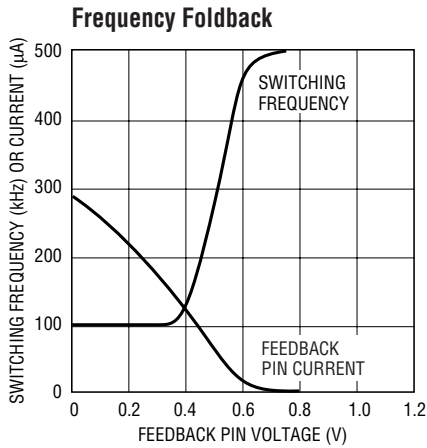
1959 G08

Error Amplifier Transconductance

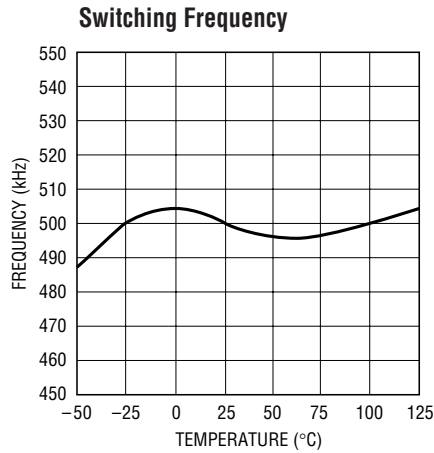


1959 G09

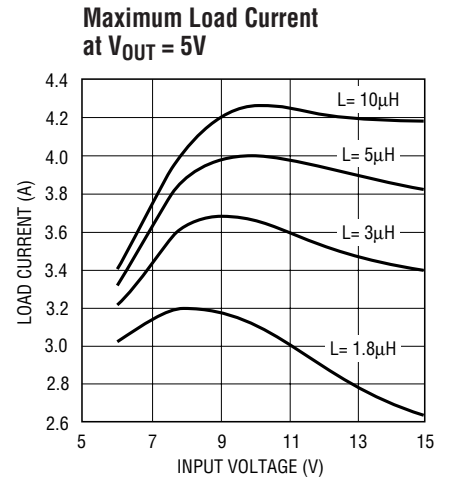
TYPICAL PERFORMANCE CHARACTERISTICS



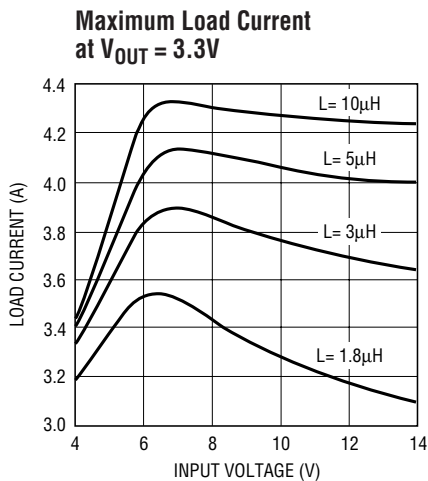
1959 G10



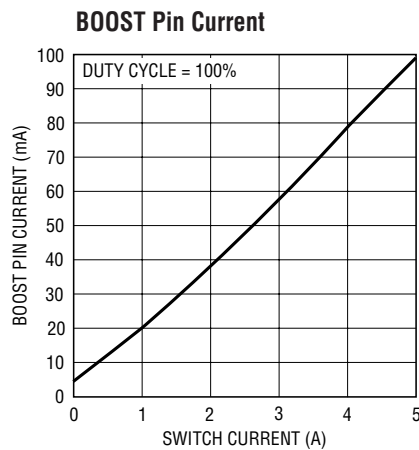
1959 G11



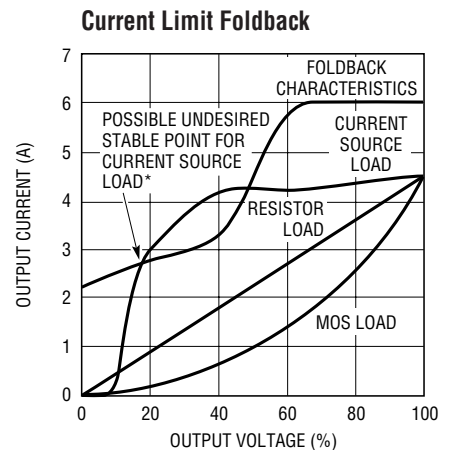
1959 G13



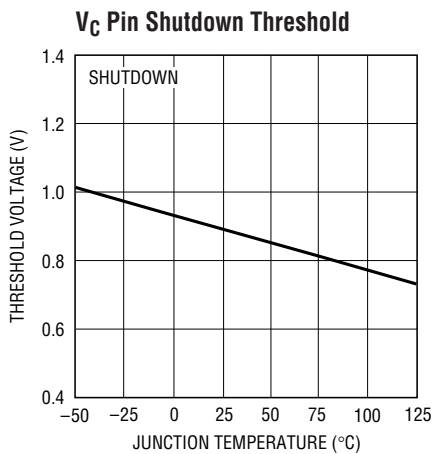
1959 G14



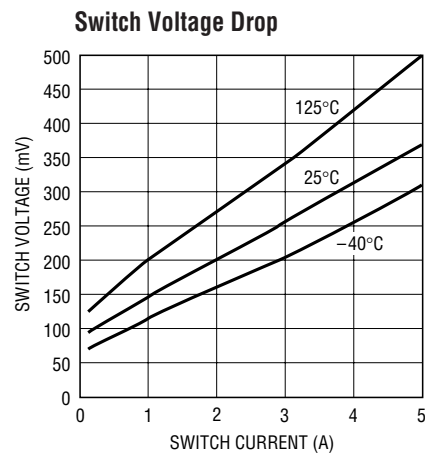
1959 G15



1959 G16



1959 G17



1959 G18

*See "More Than Just Voltage Feedback" in the Applications Information section.

PIN FUNCTIONS

FB: The feedback pin is used to set output voltage using an external voltage divider that generates 1.21V at the pin with the desired output voltage. Three additional functions are performed by the FB pin. When the pin voltage drops below 0.8V, switch current limit is reduced. Below 0.7V the external sync function is disabled and switching frequency is reduced. See Feedback Pin Function section in Applications Information for details.

BOOST: The BOOST pin is used to provide a drive voltage, higher than the input voltage, to the internal bipolar NPN power switch. Without this added voltage, the typical switch voltage loss would be about 1.5V. The additional boost voltage allows the switch to saturate and voltage loss approximates that of a 0.07Ω FET structure, but with much smaller die area. Efficiency improves from 75% for conventional bipolar designs to > 89% for these new parts.

V_{IN}: This is the collector of the on-chip power NPN switch. This pin powers the internal circuitry and internal regulator. At NPN switch on and off, high dI/dt edges occur on this pin. Keep the external bypass and catch diode close to this pin. All trace inductance on this path will create a voltage spike at switch off, adding to the V_{CE} voltage across the internal NPN.

GND: The GND pin connection needs consideration for two reasons. First, it acts as the reference for the regulated output, so load regulation will suffer if the “ground” end of the load is not at the same voltage as the GND pin of the IC. This condition will occur when load current or other currents flow through metal paths between the GND pin and the load ground point. Keep the ground path short between the GND pin and the load and use a ground plane when possible. The second consideration is EMI caused by GND pin current spikes. Internal capacitance between the V_{SW} pin and the GND pin creates very narrow (<10ns) current spikes in the GND pin. If the GND pin is connected to system ground with a long metal trace, this trace may

radiate excess EMI. Keep the path between the input bypass and the GND pin short. The GND pin of the SO-8 package is directly attached to the internal tab. This pin should be attached to a large copper area to improve thermal resistance.

V_{SW}: The switch pin is the emitter of the on-chip power NPN switch. This pin is driven up to the input pin voltage during switch on time. Inductor current drives the switch pin negative during switch off time. Negative voltage is clamped with the external catch diode. Maximum negative switch voltage allowed is -0.8V.

SYNC: (SO8 Package Only) The sync pin is used to synchronize the internal oscillator to an external signal. It is directly logic compatible and can be driven with any signal between 10% and 90% duty cycle. The synchronizing range is equal to *initial* operating frequency, up to 1MHz. See Synchronizing section in Applications Information for details. When not in use, this pin should be grounded.

SHDN: The shutdown pin is used to turn off the regulator and to reduce input drain current to a few microamperes. Actually, this pin has two separate thresholds, one at 2.38V to disable switching, and a second at 0.4V to force complete micropower shutdown. The 2.38V threshold functions as an accurate undervoltage lockout (UVLO). This is sometimes used to prevent the regulator from operating until the input voltage has reached a predetermined level.

V_C: The V_C pin is the output of the error amplifier and the input of the peak switch current comparator. It is normally used for frequency compensation, but can do double duty as a current clamp or control loop override. This pin sits at about 1V for very light loads and 2V at maximum load. It can be driven to ground to shut off the regulator, but if driven high, current must be limited to 4mA.

BLOCK DIAGRAM

The LT1959 is a constant frequency, current mode buck converter. This means that there is an internal clock and two feedback loops that control the duty cycle of the power switch. In addition to the normal error amplifier, there is a current sense amplifier that monitors switch current on a cycle-by-cycle basis. A switch cycle starts with an oscillator pulse which sets the R_S flip-flop to turn the switch on. When switch current reaches a level set by the inverting input of the comparator, the flip-flop is reset and the switch turns off. Output voltage control is obtained by using the output of the error amplifier to set the switch current trip point. This technique means that the error amplifier commands current to be delivered to the output rather than voltage. A voltage fed system will have low phase shift up to the resonant frequency of the inductor

and output capacitor, then an abrupt 180° shift will occur. The current fed system will have 90° phase shift at a much lower frequency, but will not have the additional 90° shift until well beyond the LC resonant frequency. This makes it much easier to frequency compensate the feedback loop and also gives much quicker transient response.

High switch efficiency is attained by using the BOOST pin to provide a voltage to the switch driver which is higher than the input voltage, allowing switch to be saturated. This boosted voltage is generated with an external capacitor and diode. Two comparators are connected to the shutdown pin. One has a 2.38V threshold for undervoltage lockout and the second has a 0.4V threshold for complete shutdown.

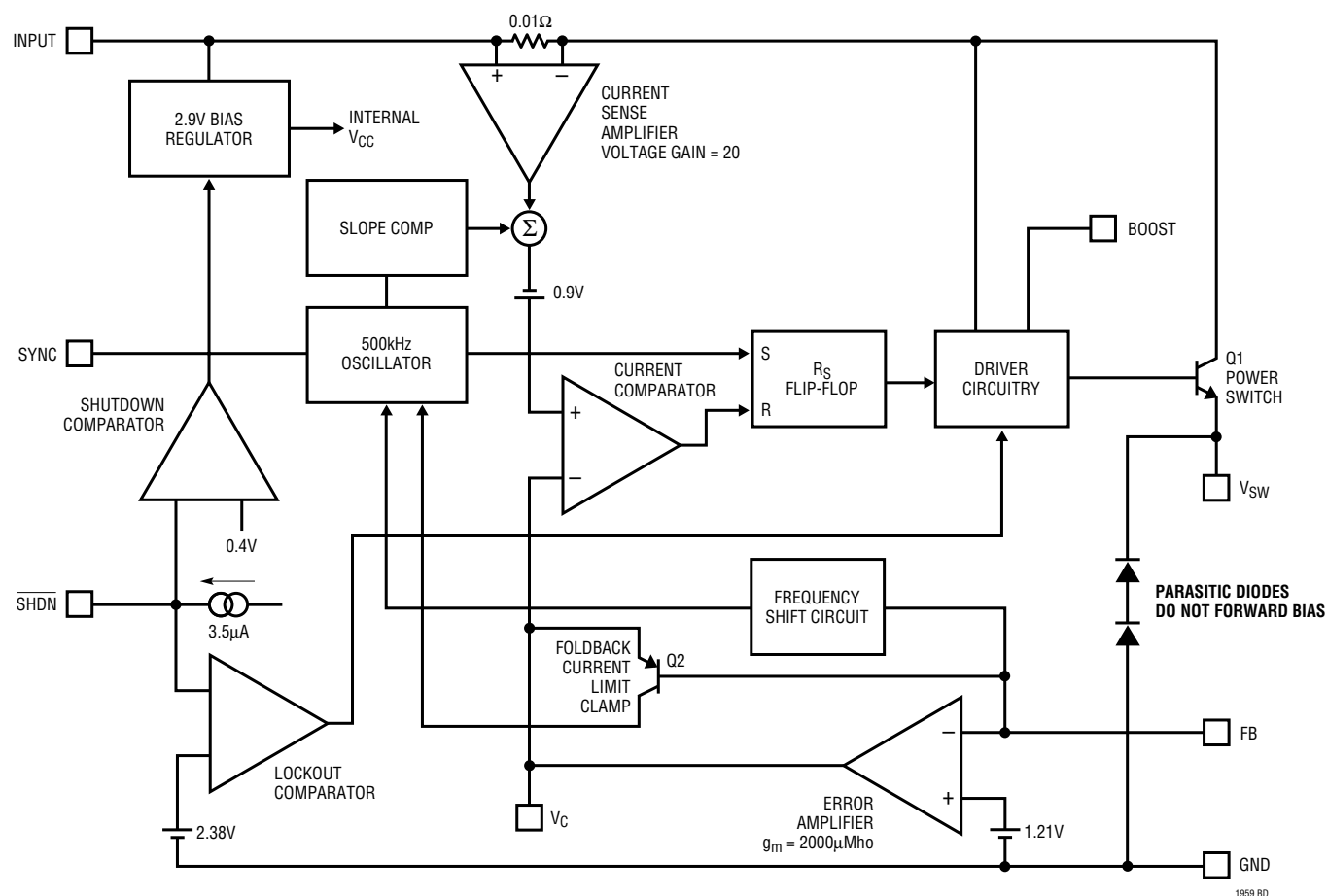


Figure 1. Block Diagram

The feedback (FB) pin on the LT1959 is used to set output voltage and provide several overload protection features. The first part of this section deals with selecting resistors to set output voltage and the remaining part talks about foldback frequency and current limiting created by the FB pin. Please read both parts before committing to a final design.

$$R1 = \frac{R2(V_{OUT} - 1.21)}{1.21}$$


The feedback pin is used for more than just output voltage sensing. It also reduces switching frequency and current limit when output voltage is very low (see the Frequency Foldback graph in Typical Performance Characteristics). This is done to control power dissipation in both the IC and in the external diode and inductor during short-circuit conditions. A shorted output requires the switching regulator to operate at very low duty cycles, and the average current through the diode and inductor is equal to the short-circuit current limit of the switch (typically 6A for the LT1959, folding back to less than 3A). Minimum switch on time limitations would prevent the switcher from attaining a sufficiently low duty cycle if switching frequency were maintained at 500kHz, so frequency is reduced by about 5:1 when the feedback pin voltage drops below 0.5V (see Frequency Foldback graph). This does not affect operation with normal load conditions; one simply sees a gear shift in switching frequency during start-up as the output voltage rises.

In addition to lower switching frequency, the LT1959 also operates at lower switch current limit when the feedback pin voltage drops below 0.8V. Q2 in Figure 2 performs this function by clamping the V_C pin to a voltage less than its normal 2.1V upper clamp level. This *foldback current limit* greatly reduces power dissipation in the IC, diode and inductor during short-circuit conditions. External synchronization is also disabled to prevent interference with foldback operation. Again, it is nearly transparent to the user under normal load conditions. The only loads that may be affected are current source loads which maintain full load current with output voltage less than 50% of final value. In these rare situations the feedback pin can be clamped above 0.75V with an external diode to defeat foldback current limit. *Caution:* clamping the feedback pin means that frequency shifting will also be defeated, so a combination of high input voltage and dead shorted output may cause the LT1959 to lose control of current limit.

APPLICATIONS INFORMATION

The internal circuitry which forces reduced switching frequency also causes current to flow out of the feedback pin when output voltage is low. The equivalent circuitry is shown in Figure 2. Q1 is completely off during normal operation. If the FB pin falls below 0.7V, Q1 begins to conduct current and reduces frequency at the rate of approximately 2kHz/μA. To ensure adequate frequency foldback (under worst-case short-circuit conditions), the external divider Thevinin resistance must be low enough to pull 150μA out of the FB pin with 0.3V on the pin ($R_{DIV} \leq 2k$). *The net result is that reductions in frequency and current limit are affected by output voltage divider impedance. Although divider impedance is not critical, caution should be used if resistors are increased beyond the suggested values and short-circuit conditions will occur with high input voltage.* High frequency pickup will increase and the protection accorded by frequency and current foldback will decrease.

MAXIMUM OUTPUT LOAD CURRENT

Maximum load current for a buck converter is limited by the maximum switch current rating (I_P) of the LT1959. This current rating is 4.5A up to 50% duty cycle (DC), decreasing to 3.7A at 80% duty cycle. This is shown graphically in Typical Performance Characteristics and as shown in the formula below:

$$I_P = 4.5A \text{ for } DC \leq 50\%$$

$$I_P = 3.21 + 5.95(DC) - 6.75(DC)^2 \text{ for } 50\% < DC < 90\%$$

$$DC = \text{Duty cycle} = V_{OUT}/V_{IN}$$

Example: with $V_{OUT} = 5V$, $V_{IN} = 8V$; $DC = 5/8 = 0.625$, and;

$$I_{SW(MAX)} = 3.21 + 5.95(0.625) - 6.75(0.625)^2 = 4.3A$$

Current rating decreases with duty cycle because the LT1959 has internal slope compensation to prevent current mode subharmonic switching. For more details, read Application Note 19. The LT1959 is a little unusual in this regard because it has nonlinear slope compensation which gives better compensation with less reduction in current limit.

Maximum load current would be equal to maximum switch current *for an infinitely large inductor*, but with

finite inductor size, maximum load current is reduced by one-half peak-to-peak inductor current. The following formula assumes continuous mode operation, implying that the term on the right is less than one-half of I_P .

$$I_{OUT(MAX)} = I_P - \frac{(V_{OUT})(V_{IN} - V_{OUT})}{2(L)(f)(V_{IN})}$$

For the conditions above and $L = 3.3\mu H$,

$$\begin{aligned} I_{OUT(MAX)} &= 4.3 - \frac{(5)(8-5)}{2(3.3 \cdot 10^{-6})(500 \cdot 10^3)(8)} \\ &= 4.3 - 0.57 = 3.73A \end{aligned}$$

At $V_{IN} = 15V$, duty cycle is 33%, so I_P is just equal to a fixed 4.5A, and $I_{OUT(MAX)}$ is equal to:

$$\begin{aligned} &4.5 - \frac{(5)(15-5)}{2(3.3 \cdot 10^{-6})(500 \cdot 10^3)(15)} \\ &= 4.5 - 1.01 = 3.49A \end{aligned}$$

Note that there is less load current available at the higher input voltage because inductor ripple current increases. This is not always the case. Certain combinations of inductor value and input voltage range may yield lower available load current at the lowest input voltage due to reduced peak switch current at high duty cycles. If load current is close to the maximum available, please check maximum available current at both input voltage extremes. To calculate actual peak switch current with a given set of conditions, use:

$$I_{SW(PEAK)} = I_{OUT} + \frac{V_{OUT}(V_{IN} - V_{OUT})}{2(L)(f)(V_{IN})}$$

APPLICATIONS INFORMATION

CHOOSING THE INDUCTOR AND OUTPUT CAPACITOR

For most applications the output inductor will fall in the range of 3μH to 20μH. Lower values are chosen to reduce physical size of the inductor. Higher values allow more output current because they reduce peak current seen by the LT1959 switch, which has a 4.5A limit. Higher values also reduce output ripple voltage, and reduce core loss. Graphs in the Typical Performance Characteristics section show maximum output load current versus inductor size and input voltage.

When choosing an inductor you might have to consider maximum load current, core and copper losses, allowable component height, output voltage ripple, EMI, fault current in the inductor, saturation, and of course, cost. The following procedure is suggested as a way of handling these somewhat complicated and conflicting requirements.

1. Choose a value in microhenries from the graphs of maximum load current and core loss. Choosing a small inductor with lighter loads may result in discontinuous mode of operation, but the LT1959 is designed to work well in either mode. Keep in mind that lower core loss means higher cost, at least for closed core geometries like toroids. The core loss graphs show absolute loss for a 3.3V output, so actual percent losses must be calculated for each situation.

Assume that the average inductor current is equal to load current and decide whether or not the inductor must withstand continuous fault conditions. If maximum load current is 0.5A, for instance, a 0.5A inductor may not survive a continuous 4.5A overload condition. Dead shorts will actually be more gentle on the inductor because the LT1959 has foldback current limiting.

2. Calculate peak inductor current at full load current to ensure that the inductor will not saturate. Peak current can be significantly higher than output current, especially with smaller inductors and lighter loads, so don't omit this step. Powdered iron cores are forgiving because they saturate softly, whereas ferrite cores

saturate abruptly. Other core materials fall in between somewhere. The following formula assumes continuous mode of operation, but it errs only slightly on the high side for discontinuous mode, so it can be used for all conditions.

$$I_{\text{PEAK}} = I_{\text{OUT}} + \frac{V_{\text{OUT}}(V_{\text{IN}} - V_{\text{OUT}})}{2(f)(L)(V_{\text{IN}})}$$

V_{IN} = Maximum input voltage

f = Switching frequency, 500kHz

3. Decide if the design can tolerate an "open" core geometry like a rod or barrel, which have high magnetic field radiation, or whether it needs a closed core like a toroid to prevent EMI problems. One would not want an open core next to a magnetic storage media, for instance! This is a tough decision because the rods or barrels are temptingly cheap and small and there are no helpful guidelines to calculate when the magnetic field radiation will be a problem.
4. Start shopping for an inductor (see representative surface mount units in Table 2) which meets the requirements of core shape, peak current (to avoid saturation), average current (to limit heating), and fault current (if the inductor gets too hot, wire insulation will melt and cause turn-to-turn shorts). Keep in mind that all good things like high efficiency, low profile, and high temperature operation will increase cost, sometimes dramatically. Get a quote on the cheapest unit first to calibrate yourself on price, then ask for what you really want.
5. After making an initial choice, consider the secondary things like output voltage ripple, second sourcing, etc. Use the experts in the Linear Technology's applications department if you feel uncertain about the final choice. They have experience with a wide range of inductor types and can tell you about the latest developments in low profile, surface mounting, etc.

APPLICATIONS INFORMATION

Table 2

VENDOR/ PART NO.	VALUE (μ H)	DC (Amps)	CORE TYPE	SERIES RESIS- TANCE(Ω)	CORE MATER- IAL	HEIGHT (mm)
Coiltronics						
CTX2-1	2	4.1	Tor	0.011	KM μ	4.2
CTX5-4	5	4.4	Tor	0.019	KM μ	6.4
CTX8-4	8	3.5	Tor	0.020	KM μ	6.4
CTX2-1P	2	3.4	Tor	0.014	52	4.2
CTX2-3P	2	4.6	Tor	0.012	52	4.8
CTX5-4P	5	3.3	Tor	0.027	52	6.4
Sumida						
CDRH125	10	4.0	SC	0.025	Fer	6
CDRH125	12	3.5	SC	0.027	Fer	6
CDRH125	15	3.3	SC	0.030	Fer	6
CDRH125	18	3.0	SC	0.034	Fer	6
Coilcraft						
DT3316-222	2.2	5	SC	0.035	Fer	5.1
DT3316-332	3.3	5	SC	0.040	Fer	5.1
DT3316-472	4.7	3	SC	0.045	Fer	5.1
Pulse						
PE-53650	4	4.8	Tor	0.017	52	9.1
PE-53651	5	5.4	Tor	0.018	52	9.1
PE-53652	9	5.5	Tor	0.022	52	10
PE-53653	16	5.1	Tor	0.032	52	10
Dale						
IHSM-4825	2.7	5.1	Open	0.034	Fer	5.6
IHSM-4825	4.7	4.0	Open	0.047	Fer	5.6
IHSM-5832	10	4.3	Open	0.053	Fer	7.1
IHSM-5832	15	3.5	Open	0.078	Fer	7.1
IHSM-7832	22	3.8	Open	0.054	Fer	7.1

Tor = Toroid

SC = Semiclosed geometry

Fer = Ferrite core material

52 = Type 52 powdered iron core material

KM μ = Kool M μ ®

Output Capacitor

The output capacitor is normally chosen by its Effective Series Resistance (ESR), because this is what determines output ripple voltage. At 500kHz, any polarized capacitor is essentially resistive. To get low ESR takes *volume*, so physically smaller capacitors have high ESR. The ESR

range for typical LT1959 applications is 0.05 Ω to 0.2 Ω . A typical output capacitor is an AVX type TPS, 100 μ F at 10V, with a guaranteed ESR less than 0.1 Ω . This is a “D” size surface mount solid tantalum capacitor. TPS capacitors are specially constructed and tested for low ESR, so they give the lowest ESR for a given volume. The value in microfarads is not particularly critical, and values from 22 μ F to greater than 500 μ F work well, but you cannot cheat mother nature on ESR. If you find a tiny 22 μ F solid tantalum capacitor, it will have high ESR, and output ripple voltage will be terrible. Table 3 shows some typical solid tantalum surface mount capacitors.

Table 3. Surface Mount Solid Tantalum Capacitor ESR and Ripple Current

E Case Size	ESR (Max., Ω)	Ripple Current (A)
AVX TPS, Sprague 593D	0.1 to 0.3	0.7 to 1.1
AVX TAJ	0.7 to 0.9	0.4
D Case Size		
AVX TPS, Sprague 593D	0.1 to 0.3	0.7 to 1.1
C Case Size		
AVX TPS	0.2 (typ)	0.5 (typ)

Many engineers have heard that solid tantalum capacitors are prone to failure if they undergo high surge currents. This is historically true, and type TPS capacitors are specially tested for surge capability, but surge ruggedness is not a critical issue with the *output* capacitor. Solid tantalum capacitors fail during very high *turn-on* surges, which do not occur at the output of regulators. High *discharge* surges, such as when the regulator output is dead shorted, do not harm the capacitors.

Unlike the input capacitor, RMS ripple current in the output capacitor is normally low enough that ripple current rating is not an issue. The current waveform is triangular with a typical value of 200mA_{RMS}. The formula to calculate this is:

Output Capacitor Ripple Current (RMS):

$$I_{\text{RIPPLE(RMS)}} = \frac{0.29(V_{\text{OUT}})(V_{\text{IN}} - V_{\text{OUT}})}{(L)(f)(V_{\text{IN}})}$$

Kool M μ is a registered trademark of Magnetics, Inc.

APPLICATIONS INFORMATION

Ceramic Capacitors

Higher value, lower cost ceramic capacitors are now available in smaller case sizes. These are ideal for input bypassing because of their high ripple rating and tolerance to turn-on surges. As output capacitors, caution must be used. Solid tantalum capacitor's ESR generates a loop "zero" at 5kHz to 50kHz that is beneficial in giving acceptable loop phase margin. Ceramic capacitors remain capacitive to beyond 300kHz and usually resonate with their ESL before ESR becomes effective. When using ceramic output capacitors, the loop compensation pole frequency must be reduced by a typical factor of 10.

OUTPUT RIPPLE VOLTAGE

Figure 3 shows a typical output ripple voltage waveform for the LT1959. Ripple voltage is determined by the high frequency impedance of the output capacitor, and ripple current through the inductor. Peak-to-peak ripple current through the inductor into the output capacitor is:

$$I_{P-P} = \frac{(V_{OUT})(V_{IN} - V_{OUT})}{(V_{IN})(L)(f)}$$

For high frequency switchers, the sum of ripple current slew rates may also be relevant and can be calculated from:

$$\Sigma \frac{dI}{dt} = \frac{V_{IN}}{L}$$

Peak-to-peak output ripple voltage is the sum of a *triwave* created by peak-to-peak ripple current times ESR, and a *square* wave created by parasitic inductance (ESL) and ripple current slew rate. Capacitive reactance is assumed to be small compared to ESR or ESL.

$$V_{RIPPLE} = (I_{P-P})(ESR) + (ESL) \Sigma \frac{dI}{dt}$$

Example: with $V_{IN}=10V$, $V_{OUT}=5V$, $L=10\mu H$, $ESR=0.1\Omega$, $ESL=10nH$:

$$I_{P-P} = \frac{(5)(10-5)}{(10)(10 \cdot 10^{-6})(500 \cdot 10^3)} = 0.5A$$

$$\Sigma \frac{dI}{dt} = \frac{10}{10 \cdot 10^{-6}} = 10^6$$

$$V_{RIPPLE} = (0.5A)(0.1) + (10 \cdot 10^{-9})(10^6)$$

$$= 0.05 + 0.01 = 60mV_{P-P}$$

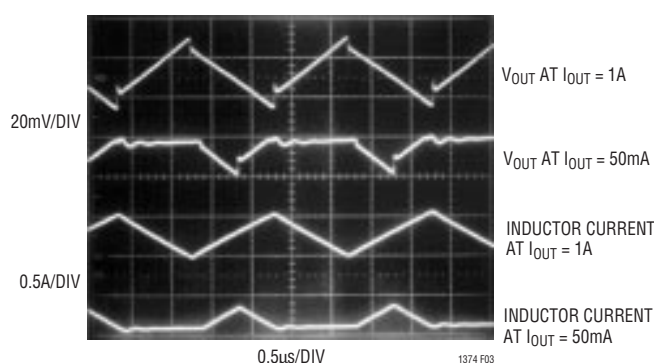


Figure 3. LT1959 Ripple Voltage Waveform

CATCH DIODE

The suggested catch diode (D1) is a 1N5821 Schottky, or its Motorola equivalent, MBR330. It is rated at 3A average forward current and 30V reverse voltage. Typical forward voltage is 0.5V at 3A. The diode conducts current only during switch off time. Peak reverse voltage is equal to regulator input voltage. Average forward current in normal operation can be calculated from:

$$I_{D(AVG)} = \frac{I_{OUT}(V_{IN} - V_{OUT})}{V_{IN}}$$

This formula will not yield values higher than 3A with maximum load current of 4.25A unless the ratio of input to output voltage exceeds 3.4:1. The only reason to consider a larger diode is the worst-case condition of a high input voltage and *overloaded* (not shorted) output. Under short-circuit conditions, foldback current limit will reduce diode current to less than 2.6A, but if the output is overloaded

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and does not fall to less than 1/3 of nominal output voltage, foldback will not take effect. With the overloaded condition, output current will increase to a typical value of 5.7A, determined by peak switch current limit of 6A. With $V_{IN} = 15V$, $V_{OUT} = 4V$ (5V overloaded) and $I_{OUT} = 5.7A$:

$$I_{D(AVG)} = \frac{5.7(15 - 4)}{15} = 4.18A$$

This is safe for short periods of time, but it would be prudent to check with the diode manufacturer if continuous operation under these conditions must be tolerated.

BOOST PIN CONSIDERATIONS

For most applications, the boost components are a 0.27 μ F capacitor and a 1N914 or 1N4148 diode. The anode is connected to the regulated output voltage and this generates a voltage across the boost capacitor nearly identical to the regulated output. In certain applications, the anode may instead be connected to the unregulated input voltage. This could be necessary if the regulated output voltage is very low (< 3V) or if the input voltage is less than 5V. Efficiency is not affected by the capacitor value, but the capacitor should have an ESR of less than 1 Ω to ensure that it can be recharged fully under the worst-case condition of minimum input voltage. Almost any type of film or ceramic capacitor will work fine.

For nearly all applications, a 0.27 μ F boost capacitor works just fine, but for the curious, more details are provided here. The size of the boost capacitor is determined by switch drive current requirements. During switch on time, drain current on the capacitor is approximately $I_{OUT}/50$. At peak load current of 4.25A, this gives a total drain of 85mA. Capacitor ripple voltage is equal to the product of on time and drain current divided by capacitor value; $\Delta V = (t_{ON})(85mA/C)$. To keep capacitor ripple voltage to less than 0.6V (a slightly arbitrary number) at the worst-case condition of $t_{ON} = 1.8\mu s$, the capacitor needs to be 0.27 μ F. Boost capacitor ripple voltage is not a critical parameter, but if the minimum voltage across the capacitor drops to less than 3V, the power switch may not saturate fully and efficiency will drop. An *approximate* formula for absolute minimum capacitor value is:

$$C_{MIN} = \frac{(I_{OUT}/50)(V_{OUT}/V_{IN})}{(f)(V_{OUT} - 2.8V)}$$

f = Switching frequency

V_{OUT} = Regulated output voltage

V_{IN} = Minimum input voltage

This formula can yield capacitor values substantially less than 0.27 μ F, but it should be used with caution since it does not take into account secondary factors such as capacitor series resistance, capacitance shift with temperature and output overload.

SHUTDOWN FUNCTION AND UNDERVOLTAGE LOCKOUT

Figure 4 shows how to add undervoltage lockout (UVLO) to the LT1959. Typically, ULVO is used in situations where the input supply is *current limited*, or has a relatively high source resistance. A switching regulator draws constant power from the source, so source current increases as source voltage drops. This looks like a negative resistance load to the source and can cause the source to current limit or latch low under low source voltage conditions. ULVO prevents the regulator from operating at source voltages where these problems might occur.

Threshold voltage for lockout is about 2.38V. A 3.5 μ A bias current flows *out* of the pin at threshold. This internally generated current is used to force a default high state on the shutdown pin if the pin is left open. When low shutdown current is not an issue, the error due to this current can be minimized by making R_{LO} 10k or less. If shutdown current is an issue, R_{LO} can be raised to 100k, but the error due to initial bias current and changes with temperature should be considered.

$$R_{LO} = 10k \text{ to } 100k \text{ (25k suggested)}$$

$$R_{HI} = \frac{R_{LO}(V_{IN} - 2.38V)}{2.38V - R_{LO}(3.5\mu A)}$$

V_{IN} = Minimum input voltage